

# ARE WE CHASING SHADOWS? QUANTIFYING UNATTRIBUTABLE POLARIZATION AND ATTRIBUTING THE REST TO ANNOTATOR GROUPS

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## ABSTRACT

Polarization has recently been proposed as a robust way of distinguishing minor disagreements from systematic differences in opinion for imbalanced, minority groups but there currently exists no methodology for attributing polarization to these groups. In this study, we discover and investigate two major limitations in realistic settings: (1) the presence of “inherent” polarization that cannot be attributed to any known or latent groups, and (2) opposing polarization effects canceling each other out in aggregated annotations. To address these issues, we introduce a new metric that attributes polarization to any annotator group while avoiding these limitations, tailored to smaller datasets common in NLP, and provide an open-source Python library implementation. We apply our method to four content moderation datasets and confirm that gender and ethnicity consistently explain polarization patterns, while ideology and personal experiences may be more important than previously thought. Our work overall indicates that subjective NLP datasets may need fewer items and more annotators per item.

## 1 INTRODUCTION

Annotations are essential for subjective tasks, such as content moderation, toxicity, and Hate Speech (HS) detection, which are especially important for protecting minority groups United Nations (2025); Dioum (2025). However, in practice, analysis rests on inter-annotator agreement or majority-vote Ivey et al. (2025); Kralj Novak et al. (2022); van der Velden et al. (2025), which may lead to biased and misconfigured systems (e.g., toxicity models disproportionately marking African American (AA) speech as “toxic” Sap et al. (2019)), or discard *fundamental* disagreements between minority groups and the majority Quaranto (2022), (see the example in Fig. 1).<sup>1</sup>

One alternative is using *polarization*, which correlates better with human judgments and is much less sensitive to different group sizes Pavlopoulos & Likas (2023; 2024). While we can detect polarization, there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following Research Question (RQ):

**RQ.** *How much polarization can we practically attribute to annotator subgroups given existing datasets?*

While prior work assumed that all polarization can be explained and attributed to some annotator characteristic, through the exhaustive creation of theoretical annotator groups our study discovers the existence of Inherent Polarization (InPol), which can not be attributed to *any possible group* of annotators—known or unknown (Finding 1). We also discover that polarization across comments has a *direction*, preventing us from aggregating annotations on the dataset level to circumvent annotation sparsity (Finding 2).

In order to solve these issues we introduce a new metric—“*apunim*”—which attributes polarization to specific annotator sociodemographic and ideological characteristics (referred to as Personal Char-

<sup>1</sup>The simulated experiments in this study were implemented by generating random annotations following the normal distribution with differing means and Standard Deviation (SD)—see App. F.

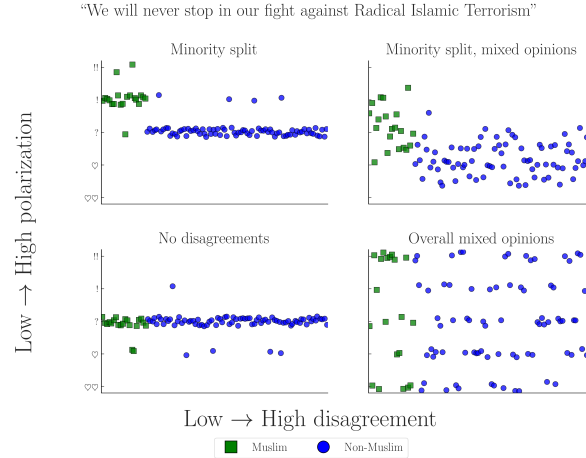


Figure 1: Simulated experiment exploring four scenarios of a HS detection task applied to a controversial statement by D. Trump (Jan. 2019). The variance in annotations across different annotators may cause traditional agreement metrics to ignore minority opinions, especially when opinions are mixed (top-right).

acteristics (PCs) in this study). Our metric is generalizable across any domain and modality (e.g., video/image classification) and, unlike previous approaches, enables both *quantitative* comparisons across datasets and statistical analysis via p-value estimation. Even in cases where unattributable polarization exists, *apunim* is capable of detecting general patterns of polarization across multiple groups. Additionally, our metric is designed to be used for existing Natural Language Processing (NLP) datasets which are usually small w.r.t. both content and annotations due to budget constraints Chen et al. (2022); Golazizian et al. (2024).

We then apply our metric to four NLP datasets focused on Toxicity and HS detection (§6), finding that annotator gender and ethnicity significantly impact annotation (Finding 4). Our findings also indicate that ideological attributes and the annotators’ personal experiences may play a major role in explaining polarization (e.g., the more religious the annotators, the more polarized they become—Finding 5), a possibility not adequately explored by relevant literature. Finally, we empirically investigate and discuss the applicability of our methodology to the modern NLP dataset landscape (App. I), and release an open source Python (PyPi) library that implements our metric.<sup>2</sup>

**Warning: This publication includes examples of controversial and potentially harmful speech for the purposes of demonstration. These examples do not represent the views of the authors.**

## 2 BACKGROUND AND RELATED WORK

### 2.1 AGREEMENT

Popular inter-annotator agreement metrics such as Cohen’s kappa, Krippendorff’s alpha, and Fleiss’s kappa aim to distinguish genuine disagreement from random disagreement through statistical chance-correction. However, these metrics are often difficult to apply to intergroup agreement without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds (Feinstein & Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), producing results that are difficult to compare across studies, being sensitive to class imbalance (Feinstein & Cicchetti, 1990), and relying on questionable assumptions about chance agreement Checco et al. (2017). While recent work has addressed some of these limitations with methods that improve applicability and robustness van der Velden et al. (2025); Wong et al.

<sup>2</sup>Library: <https://anonymous.4open.science/r/apunim-5F70>, Webpage: <https://anonymous.4open.science/r/apunim-page>, Replication code: <https://anonymous.4open.science/r/aposteriori-unimodality-E8C1>

| Dataset   | #Com | #Ann  | #Dims | Task |
|-----------|------|-------|-------|------|
| Kumar     | 2998 | 5     | 10    | Tox  |
| Sap       | 585  | 5–6   | 3     | HS   |
| DICES-350 | 350  | 123   | 4     | HS   |
| DICES-990 | 990  | 70–76 | 4     | HS   |

Table 1: Datasets used in this study. *#Ann* refers to the number of annotations per comment (see App. B), and *#Dims* to PC dimensions (e.g., age, gender). Tasks are either Toxicity (Tox) or HS.

(2021); Ivey et al. (2025), they are much more complex and less intuitive compared to their traditional counterparts, indicating that agreement may not be the correct tool for analysis of minority viewpoints.

## 2.2 POLARIZATION

There are multiple competing formulations for polarization. One of these approaches Akhtar et al. (2019) utilizes  $\chi^2$  to estimate in-group vs out-group variance, but still requires traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric Checco et al. (2017), or non-parametric Pavlopoulos & Likas (2024); Mignemi et al. (2024) approaches to the annotation histograms.

For our work, we use a non-parametric polarization metric, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics (Pavlopoulos & Likas, 2024). nDFU takes values in the range  $[0, 1]$ , where values close to 0 indicate no polarization (i.e., opinions are concentrated around a single mode). The metric is defined as follows:

$$\text{nDFU}(X) = \frac{\max_{i \neq m} \left( f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m} \quad (1)$$

$$m = \arg \max_i f_i$$

where,  $X$  denotes the set of annotations,  $f_i$  the relative frequency of the  $i$ -th annotation (e.g., the frequency of 2 in a scale of 1–5), and  $m$  the histogram peak. Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak (see Fig. 2, left for an example).

## 2.3 ATTRIBUTING POLARIZATION TO GROUPS

While many studies study the effect of subgroups on annotation agreement Goyal et al. (2022); Binns et al. (2017); Giorgi et al. (2025); Al Kuwatly et al. (2020); Sachdeva et al. (2022), attributing *polarization* is a relatively new concept. Under the name *a posteriori unimodality* Pavlopoulos & Likas (2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then it is explained by those groups. Though promising, this mechanism operates on the assumption that *all* polarization can be explained by a single split, which we disprove prove to be impossible in many cases (§4). Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.<sup>3</sup>

## 3 DATASETS

We use four datasets from current NLP research that include PC information at the level of individual annotators; specifically, the “Kumar” Kumar et al. (2021) and “Sap” (“Breadth of Posts”) Sap et al.

<sup>3</sup>For example, when presented with noisy estimates of a metric in the  $[0, 1]$  range, we would ideally want to distinguish between cases where attribution is close to .001, and cases where attribution is .49.

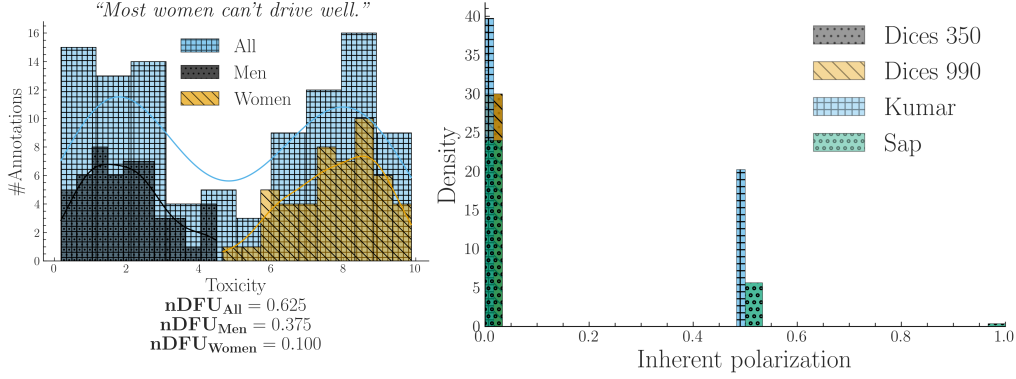


Figure 2: **Left:** Simulated experiment describing a polarizing comment where male and female annotators agree between themselves, but disagree with the opposite gender. **Right:** InPol across our datasets (§3). Contrary to assumptions made in prior work, InPol is commonly non-zero and can even reach  $nDFU = 1$  in datasets with low annotator counts.

(2022) datasets, which are concerned with Toxicity and HS detection respectively, and the DICES-350 and DICES-990 datasets Aroyo et al. (2023), which focus on LLM response safety.<sup>4</sup>

A brief overview of dataset characteristics is provided in Table 1. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct PC dimensions.<sup>5</sup> In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. More details on the annotations, including overall levels of observed polarization, can be found in App. B.

## 4 IS IT POSSIBLE TO ATTRIBUTE ALL POLARIZATION TO ANNOTATOR GROUPS?

### 4.1 PROBLEM FORMULATION

Let  $d = \{c_1, c_2, \dots\}$  be a dataset composed of multiple annotated comments.<sup>6</sup> Each comment  $c$  is assigned multiple annotators, which creates the annotation multi-set  $A(c) = \{(a_1, g_1), (a_2, g_2), \dots\}$ , where  $a_i$  is a single annotation (e.g., 2 in a LIKERT scale of 1–5), and  $g_i$  a group (e.g., men). As an example, the annotations for a comment in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\text{"Could be better, could be worse"}) = \{ (+, F), (-, M), (+, F), \dots \}$$

Intuitively, a group partially explains the polarization of a comment  $c$  when the annotations of that group  $A(c|g) = \{a(c, g') \in A(c) \mid g' = g\}$  exhibit higher polarization compared to the full set of annotations  $A(c)$ . Consider the case where a misogynistic comment is annotated for toxicity by male and female annotators shown in Fig. 2, left. The annotations are generally polarized ( $nDFU_{all} = .62$ ); but both female ( $nDFU_{women} = .10$ ) and male ( $nDFU_{men} = .37$ ) annotators exhibit low polarization between their respective groups. This pattern suggests that the overall polarization is partially caused by substantive disagreements between male and female annotators.<sup>7</sup>

<sup>4</sup>For the latter, we consider only labels relating to racism (Q3\_bias\_overall), thereby transforming the task to HS detection.

<sup>5</sup>We sample 3,000 comments in this study due to computational constraints (App. E). We run ablations for both different and differently sized-samples. Our results remain consistent, although large samples (10,000+) tend to produce null results (App. H).

<sup>6</sup>The annotated items could be of any modality, such images, videos or survey questions, even though we only consider comments in this study.

<sup>7</sup>Since polarization persists among male annotators, another subgroup of men may also contribute to it.

## 4.2 INHERENT POLARIZATION

### 4.2.1 WHAT IS IT?

Inherent Polarization (InPol) refers to disagreement that cannot be attributed to *any specific* grouping of annotators; that is, polarization that exists no matter how we split the annotators in any arbitrary group. Prior work Pavlopoulos & Likas (2024) attributed polarization to annotator groups only when the observed polarization for each individual group became exactly zero (e.g.  $nDFU(all) > nDFU(men) = nDFU(women) = 0$ ). The presence of InPol would invalidate this theory, since there would be no possible group or combination of groups  $g$  s.t.  $nDFU(g) = 0$ .

**Definition 1** (Inherent Polarization). Polarization that cannot be explained by any known or latent annotator grouping.

Formally, the InPol for a single comment  $c$  is the minimum polarization exhibited by any arbitrary group:

$$Pol_{inh}(c) = \min_{S \subseteq A(c), |S| \geq 3} \{nDFU(S)\} \quad (2)$$

When considered exhaustively, these theoretical, random groups include both observed and latent groups (e.g., annotators may be less strict if they had lunch prior to annotation). Since  $nDFU$  operates only when a group has at least three annotators (Eq. 1), this yields  $\sum_{k=3}^n \binom{n}{k}$  possible groups.

### 4.2.2 DOES IT EXIST?

We systematically explore the space of annotator groups by generating random partitions for each comment. Since enumerating all subsets is computationally intractable for large annotator counts, we establish an upper bound via Monte Carlo sampling by drawing  $k$  random partitions with random sizes:

$$Pol_{inh}^{MC}(c) = \min_{j=1}^k \{nDFU(\tilde{r}_j(A(c)))\} \quad (3)$$

where  $\tilde{r}$  is an operator that selects a randomly sized, randomly selected subset. Since Eq. 3 considers a strict subset of Eq. 2, it provides an upper bound for it:

$$Pol_{inh}^{MC}(c) \geq Pol_{inh}(c) \forall c \quad (4)$$

We compute the InPol for each comment across the four datasets. For Sap and Kumar, we use the exhaustive formulation (Eq. 2), whereas for the two DICES datasets we rely on the Monte Carlo approximation with  $k = 1000$  (Eq. 3). Fig. 2, right shows that in all DICES comments there exists at least one group that fully accounts for the observed polarization.<sup>8</sup> Interestingly, this observation does not hold for the Kumar and Sap datasets, despite evaluating all possible annotator groups exhaustively. See App. G for a qualitative look through comments with high InPol.

**Finding 1.** *Comments may exhibit a degree of InPol that cannot be attributed to any known or latent annotator grouping given the annotators present in the dataset.*

## 4.3 CAN WE REMOVE IT?

Unlike disagreement, InPol does not provide any information, but rather represents noise or uncertainty translated in terms of polarization. In order to minimize it however, we need to find its cause.

**Hypothesis 1** (Low annotator count). A low number of annotators per item may provide our model with insufficient information.

**Hypothesis 2** (Inaccessible groups). InPol may arise from the presence of partially represented groups within the annotator pool. While these groups are responsible for the observed polarization, they may go undetected in our current methodology if they are represented by only one or two annotators, as these groups are discarded (§4.2).

<sup>8</sup>  $Pol_{inh}^{MC}(c) = 0 \Rightarrow Pol_{inh}(c) = 0$  due to Eq. 4. The same logic applies if we tried to increase the number of random partitions in Eq. 3.

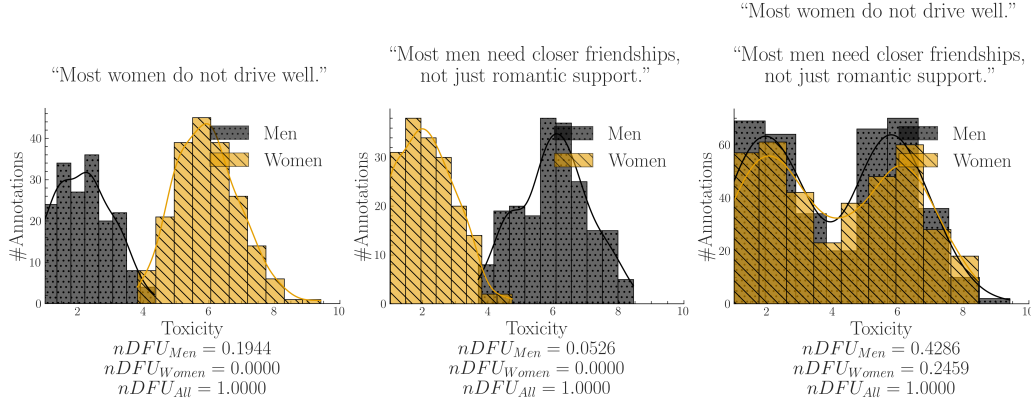


Figure 3: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization (polarization explained by gender, but not shown).

Both of these explanations are plausible and reflect known limitations of most NLP datasets. The fact that the DICES datasets (featuring 70+ annotators per item) exhibit zero InPol in all comments may initially seem to validate either of the above hypotheses. However, even when repeating the experiment using random annotator subsets (ranging from 5 to 50 annotators) and taking the maximum across ten random samples for each subset size, *we still observed zero InPol* (App. I). This finding indicates that the DICES datasets may be structurally different. This explanation is plausible given that the annotation targets are LLM texts, which may be relatively unobvious compared to human comments, as supported by low overall polarization (Fig. 6; App B).

Although a definitive answer to this question eludes us due to the lack of other datasets with high annotator counts per item, the indication that the DICES datasets may not feature similar polarization patterns is noteworthy. We discuss the exploratory use of our methodology for filtering subjective datasets in App. I.

We can not address Hyp. 2 without more annotations, but we could address Hyp. 1 by aggregating annotations between comments. However, Fig. 3 shows that two comments polarized for the opposite reason may lead to a false negative. Therefore, any statistics must be computed on the comment-level; annotations should never be aggregated across comments.<sup>9</sup>

**Finding 2.** *Items hold their own pol. direction which may compromise cross-item aggregation.*

## 5 HOW DO WE ATTRIBUTE THE REST?

### 5.1 QUANTIFYING CONTRIBUTION TO ANNOTATOR GROUPS

Instead of using InPol, which represents the minimum possible polarization exhibited by any group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups*. Specifically, we define apriori polarization for an item  $c$  according to Eq. 5, which is averaged across items for a dataset  $d$  (Eq. 6). Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by any single group.

<sup>9</sup>A similar assumption was followed in early parametric approaches, which assumed different parameters for each comment Checco et al. (2017).

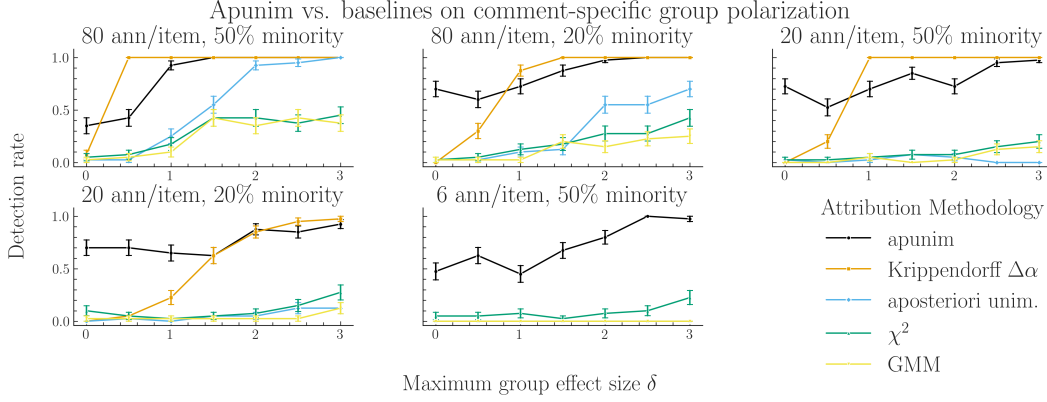


Figure 4: Polarization group attribution on randomly generated, synthetic distributions. Bars indicate standard error. For each experiment we assign a number of annotators per item, and split the annotator groups 50-50 or 80-20. Apunim is adept at attributing polarization to the correct group regardless of number of annotators per item, or how these annotators are split to different groups.

$$Pol_{apr}(c) = \frac{1}{t} \sum_{i=1}^t nDFU(\tilde{r}_i(A(c))) \quad (5) \quad Pol_{apr}(d) = \frac{1}{|d|} \sum_{c \in d} Pol_{apr}(c) \quad (6)$$

where  $\tilde{r}$  is the random partition operator (see Eq. 2). We explain why defining apriori polarization in this way is preferable to alternatives in App. C.1.

We compare this quantity with the average observed polarization of all comments (Eq. 7), which we use to compare the polarization that is attributed to a single annotator group with the expected one (Eq. 6), yielding our normalized metric, *apunim* (Eq. 8).

$$Pol_{obs}(d, g) = \frac{1}{|d|} \sum_{c \in d} nDFU(A(c | g)) \quad (7) \quad apunim(d, g) = \frac{Pol_{apr}(d) - Pol_{obs}(d, g)}{1 - Pol_{apr}(d)} \quad (8)$$

Apunim measures the degree to which an annotator group contributes to overall polarization in a set of comments. The *sign* of apunim carries information agreement-based attribution does not:

- $apunim(g) \approx 0$ : The group  $g$  does not meaningfully disagree with the other annotators.
- $apunim(g) > 0$ : The group  $g$  has fundamental disagreements with the other annotators.
- $apunim(g) < 0$ : The annotators of group  $g$  are polarized between themselves.

## 5.2 COMPARING APUNIM TO OTHER METHODS

While apunim rests on different assumptions than traditional agreement metrics, we can devise a set of experiments where the goal of both methodologies is uncovering a known polarization effect. To model polarization more realistically, we use a comment-specific group effect rather than assuming that annotators in one group are uniformly biased toward higher or lower ratings. Specifically, annotations are generated according to  $y_{ij} = b_j + g_i p_j + \epsilon_{ij}$ , where  $b_j$  represents the baseline rating for comment  $j$ ,  $g_i \in \{-\frac{1}{2}, +\frac{1}{2}\}$  indicates the annotator’s group, and  $p_j$  controls the magnitude and direction of polarization for comment  $j$ . We sample  $p_j$  uniformly from  $[-\delta, \delta]$ , such that different comments can favor different groups while  $\delta$  controls the maximum group-level separation. Thus, rather than modeling a fixed group bias, the simulation captures situations where groups systematically disagree on particular comments, with the direction and strength of disagreement varying across items but being consistent among annotators of the same group.

To determine whether an observed score is statistically significant, we construct a null distribution for each method. The null hypothesis is that the group labels are unrelated to the observed annotations. We therefore repeatedly shuffle the group labels across annotators, recompute the method’s

| Group                 | Apunim | Support |
|-----------------------|--------|---------|
| <b>DICES-350</b>      |        |         |
| Ethnicity=Asian       | -.066  | 8138    |
| Education=Other       | .021   | 2191    |
| <b>DICES-990</b>      |        |         |
| Ethnicity=AfricanAm   | -.091  | 5810    |
| Ethnicity=Latino      | .133   | 6610    |
| Ethnicity=Other       | -.074  | 24321   |
| Ethnicity=White       | .120   | 11392   |
| Age=GenZ              | -.063  | 13741   |
| Age=GenX+             | .046   | 14384   |
| Education=High school | -.075  | 7419    |
| Gender=Man            | -.027  | 32494   |
| Gender=Woman          | .024   | 33471   |
| <b>Sap</b>            |        |         |
| Ethnicity=AfricanAm   | .042   | 4       |
| Gender=Man            | -.091  | 1439    |
| Gender=Woman          | .244   | 877     |

| Group                         | Apunim | Support |
|-------------------------------|--------|---------|
| <b>Kumar</b>                  |        |         |
| A. Innate Characteristics     |        |         |
| Ethnicity=White               | -.029  | 6619    |
| Education=College             | -.034  | 2508    |
| SexOrient=Heterosexual        | -.050  | 5721    |
| IsTrans=No                    | -.079  | 1697    |
| B. Experiences and Ideologies |        |         |
| HasBeenTargeted=False         | -.048  | 6632    |
| IsParent=No                   | -.069  | 5514    |
| IsParent=Yes                  | .049   | 6216    |
| ReligionImportant=No          | -.062  | 3864    |
| ReligionImportant=Very        | .052   | 4062    |
| SeenToxicity=False            | .035   | 3050    |
| SeenToxicity=True             | -.077  | 6360    |

Table 2: Statistically significant ( $p < 0.05$ ) apunim results across the four datasets. Gray rows show Ethnicity and Gender groups. “Support” refers to number of annotations. Consult App. J for the full results.

statistic, and use the resulting scores to form the null distribution. The observed score is then compared with this distribution. See App. C.3 for the apunim p-value implementation.

Fig. 4 shows the effectiveness of apunim, the aposteriori unimodality framework proposed by Pavlopoulos & Likas (2024), the  $\chi^2$  method presented in Akhtar et al. (2019), GMM clustering used in Checco et al. (2017), and a baseline using inter-group disagreement ( $\Delta\alpha$  Krippendorff). Both the  $\chi^2$  and GMM methods generally fail at attributing polarization, which is most likely caused by opposing polarization effects (see Fig. 3) violating their assumptions on fixed mean differences between groups. Inter-group disagreement remains competitive when many annotators are available, but fail at the underperform in cases where the examined group is a minority and when a few annotators are available per item (a common case among NLP datasets—see §3). Apunim in contrast is robust both to the number of annotators per item, opposing polarization effects, and minority groups, which validates our initial motivation.

**Finding 3.** *Apunim beats established and proposed methodologies in realistic conditions.*

## 6 ATTRIBUTING REAL-LIFE POLARIZATION TO HUMAN GROUPS

Assuming a confidence level of  $\alpha = 0.05$  we test whether any of the provided PC dimensions can partially explain polarization in the annotations. Statistically significant results are presented in Table 2. See App. E for details on other parameters and execution times, and App. J for the full results.

**Ethnicity** Polarization is consistently attributed to ethnicity across all datasets, confirming prior work suggesting that it is a systematic source of annotator disagreement in these tasks Goyal et al. (2022). We note that polarization attribution does not have a consistent direction across datasets, which we believe to be caused by studies using different ethnic subgroups, allowing polarization detection on the level of PCs, but distorting subgroup attribution.

**Gender** In both DICES-990 and Sap, women are a consistent source of polarization (positive values), while men disagree among themselves (negative values). These results are consistent with previous findings regarding HS (Wojatzki et al., 2018) and toxicity annotation Binns et al. (2017). Non-transgender annotators also showed the highest degree of polarization in the Kumar dataset in absolute terms (−.788) suggesting that transgender annotators are a systematic source of annotator



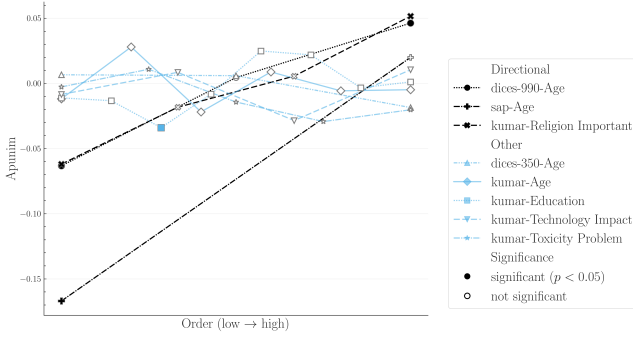


Figure 5: Polarization attribution trends in ordinal groups. The left side of the x-axis refers to young annotators, low education, low religiousness, and negativity towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in App. J. We exclude groups with  $< 50$  annotations, since they near-universally result in zero apunim values due to lack of data.

disagreement, although only the inverse effect is detectable due to annotator sparsity (hence the negative sign).

**Finding 4.** *Annotator ethnicity and gender are consistent causes of polarization.*

**Ideology and experiences** appear to be a more consistent source of annotator disagreement than inherent characteristics. Specifically, the Kumar dataset, which tracks ideological attributes, shows polarization attribution linked primarily to personal experiences and beliefs, rather than to sociodemographic dimensions, although further datasets are required to generalize this finding. The divide between ideology/personal experiences and innate sociodemographic characteristics has not been adequately explored in current literature and should be further studied in future work.

**Religion** demonstrates a clear relationship with polarization attribution. According to Fig. 5, highly devout annotators exhibit higher levels of self-agreement compared with other annotators (positive apunim). Non-religious annotators show a pattern of increased internal disagreement (negative apunim). Notably, this trend of internal disagreement gradually reverses as religious devoutness increases (increasing apunim).<sup>10</sup>

**Finding 5.** *Pol. attribution may be more pronounced in ideological attributes and personal experiences. It also escalates as annotators become more religious.*

## 7 CONCLUSION

We investigated to what extent polarization can be attributed to annotator groups. We discovered as of yet unknown issues regarding the presence of inherent and conflicting polarization, which we treat using a new metric, *apunim*. By applying this metric to four subjective NLP datasets, we discover that (1) ethnicity and gender seem to generally drive polarization attribution, and (2) ideological groups may be more important than reported in the literature. Finally, we empirically investigate and discuss how our methodology interacts with (generally small in both annotations and size) current subjective NLP datasets (App. I) and release a PyPi library implementation to support further work.

<sup>10</sup>The statistical insignificance observed during the transition from irreligious to highly religious groups may represent genuine shifts rather than inconclusive results, since Eq. 9 checks whether  $apunim \neq 0$ , given that  $apunim_{rand}(d, g)$  will usually be close to zero. This phenomenon may also apply to other ordinal PCs, such as Age in both Sap and DICES-990, but finer-grained subgroups are needed to verify this.

## LIMITATIONS

Researchers often release only aggregated single-label annotations or omit annotator PC characteristics entirely. As a result, both our analysis and the applicability of our framework are limited by the scarcity of suitable datasets. In addition, because datasets with large numbers of annotators per item are rare, our approach operates only at the dataset level. This limitation prevents us from identifying individual comments or examples that could serve as concrete explanations for the observed *apunim* values. Furthermore, our analysis does not take into account confounding variables. It is plausible that some annotator groups are correlated with others (e.g., LGBTQ+ annotators may not lean into Conservatism). This is left for future work, as the scope of this analysis is deemed too large for this paper.

## ETHICS STATEMENT

**Scope** It must be noted that the datasets utilized in this study are restricted to posts written in English and pertain exclusively to the Western, specifically U.S., cultural context. Consequently, the findings and observed phenomena are not necessarily generalizable to linguistic or cultural contexts outside of this scope, such as non-English languages or other global cultural spheres.

**Dual use concerns** The primary objective of this study is to identify systemic sources of polarization in subjective NLP datasets to facilitate the construction of more robust and equitable AI systems. However, given that our methodology allows for the quantitative attribution of specific judgment patterns to sociodemographic and ideological characteristics, we acknowledge three primary risks inherent in this research.

First, *Discriminatory Profiling*: If the metric is applied outside of a dataset quality analysis, its ability to correlate specific judgment patterns with annotator groups could be misused for surveillance or unjust profiling. Second, *Automated Bias Amplification*: The diagnostic patterns of polarization we detect are reflections of human systemic biases. Without human oversight, training subsequent models on these patterns risks automating and entrenching existing societal prejudices. Finally, the knowledge provided by *apunim* could be *weaponized*, allowing malicious actors to strategically target weak points in moderation systems or intentionally maximize societal conflict.

We stress that our metric functions strictly as a diagnostic warning signal. Any deployment in a real-world system must mandate a rigorous human-in-the-loop review, ensuring that identified patterns lead to systemic improvements (e.g., policy reform, demographic diversity initiatives) rather than merely justifying an automated system.

**Data sourcing and privacy** All datasets employed in this study were either publicly accessible or provided to us by the original authors with full disclosure regarding the purpose of our research. No identifiable, personal information was included in our pipeline. We do not release proprietary data provided by the authors.

**Examples used** Our study utilizes examples of comments that may be considered divisive across certain demographics. These examples were generated by our team solely for demonstration purposes. We explicitly state that the contents of these examples and the groups studied do not necessarily reflect the views or behaviors of the groups they represent, nor do they reflect the opinions of the authors.

**Tools used** We used small, open-source LLMs (primarily a local Gemma-4 model) to perform minor stylistic adjustments throughout this document. Furthermore, we utilized open-source and proprietary models to generate portions of the code for the plots and experiments presented. All such AI-generated outputs have been meticulously checked and validated by the authors.

## REPRODUCIBILITY STATEMENT

All code, data-processing scripts and the library implementing *apunim* are released anonymously: library <https://anonymous.4open.science/r/apunim-5F70>, documenta-

tion <https://anonymous.4open.science/r/apunim-page>, replication code <https://anonymous.4open.science/r/aposteriori-unimodality-E8C1>. Statistical procedures are detailed in App. C, replication details (seeds, compute) in App. E.

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| <b>Dataset</b> | count | mean | SD   | min | 25% | 50% | 75% | max |
|----------------|-------|------|------|-----|-----|-----|-----|-----|
| DICES-350      | 350   | 123  | 0    | 123 | 123 | 123 | 123 | 123 |
| DICES-990      | 990   | 72.8 | 1.17 | 69  | 72  | 73  | 74  | 76  |
| Sap            | 585   | 5.6  | .78  | 3   | 6   | 6   | 6   | 12  |
| Kumar          | 2998  | 5.02 | .34  | 5   | 5   | 5   | 5   | 10  |

Table 3: Number of annotations per item in each of the four datasets used in this study.

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## A ACRONYMS

**InPol** Inherent Polarization

**nDFU** normalized Distance From Unimodality

**NLP** Natural Language Processing

**GPLv3** GNU General Public Licence v3

**AA** African American

**PC** Personal Characteristic

**HS** Hate Speech

**SD** Standard Deviation

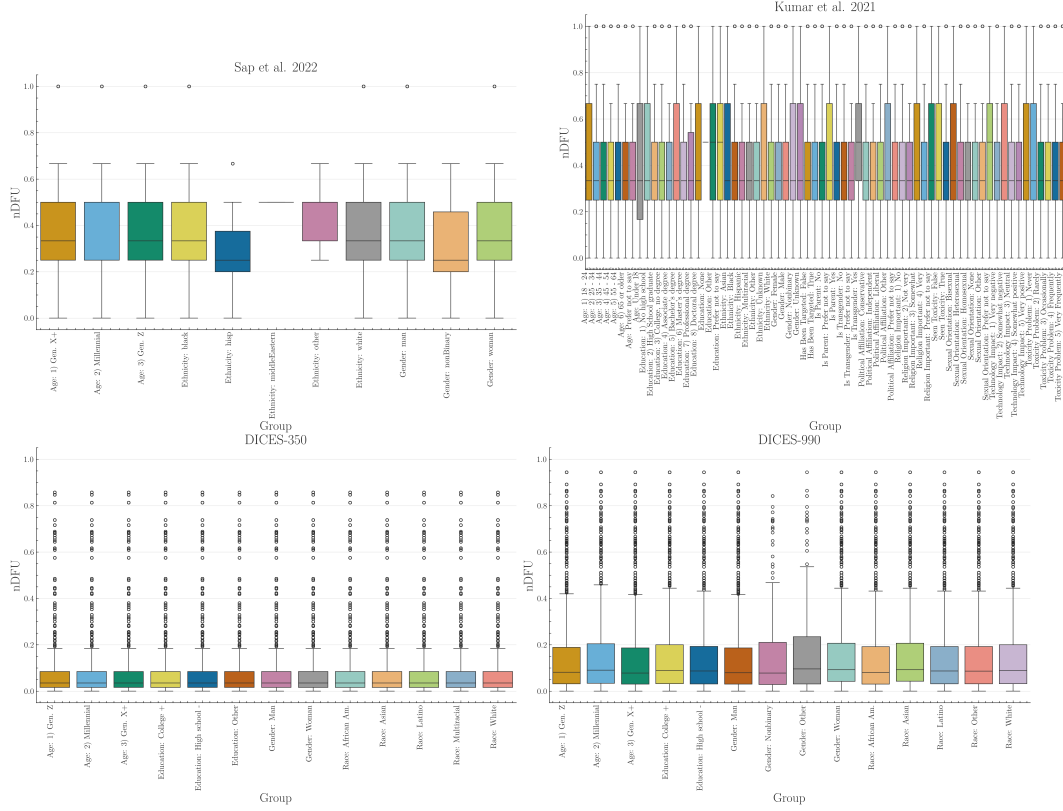


Figure 6: Boxplots of dataset polarization for the human datasets considered in this paper for all groups in each examined PC dimension.

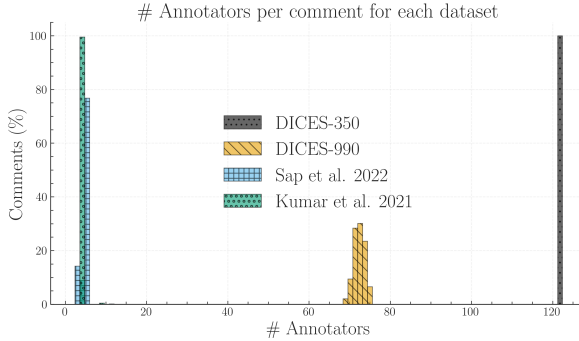


Figure 7: We calculate the ratio of comments for each dataset (y-axis) that had  $x$  number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table 3), which is likely caused by a data error.

## B ANNOTATION DETAILS

The number of annotators per comment for all datasets presented in §3 can be found in 7, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We do not include these comments, since they are very likely a data error. Tables 4,5,6,7,8 show the annotation counts for each PC group in each dataset.

| Category  | Subgroup      | Support |
|-----------|---------------|---------|
| Gender    | Woman         | 21700   |
|           | Man           | 21350   |
| Race      | White         | 10500   |
|           | African Am.   | 10150   |
|           | Asian         | 9100    |
|           | Latino        | 7700    |
|           | Multiracial   | 5600    |
| Age       | 1) Gen. Z     | 19600   |
|           | 2) Millennial | 12600   |
|           | 3) Gen. X+    | 10850   |
| Education | College +     | 26250   |
|           | High school - | 14350   |
|           | Other         | 2450    |

Table 4: Subgroup Counts for DICES-350. “Support” refers to annotation count.

| Category  | Subgroup      | Support |
|-----------|---------------|---------|
| Gender    | Woman         | 36050   |
|           | Man           | 35392   |
|           | Nonbinary     | 331     |
|           | Other         | 330     |
| Race      | Other         | 26394   |
|           | Asian         | 19885   |
|           | White         | 12322   |
|           | Latino        | 7170    |
|           | African Am.   | 6332    |
| Age       | Gen. Z        | 14899   |
|           | Millennial    | 41542   |
|           | Gen. X+       | 15662   |
| Education | College +     | 64054   |
|           | High school - | 8049    |

Table 5: Subgroup Counts for DICES-990. “Support” refers to annotation count.

## C STATISTICAL CONSIDERATIONS

### C.1 DEFINING APRIORI POLARIZATION

In §4.1, we explained how our methodology needs to compare the polarization of  $A(c)$  and  $A(c|g)$ . Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. ??, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

### C.2 FILTERING INELIGIBLE COMMENTS

Because *apunim* is fundamentally designed to quantify polarization, we exclude comments that exhibit no measurable polarization ( $nDFU(c) = 0$ ). Furthermore, we also exclude comments annotated exclusively by a single group (i.e., comments where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of nDFU

| Category  | Subgroup      | Support |
|-----------|---------------|---------|
| Age       | Gen. Z        | 252     |
|           | Millennial    | 2259    |
|           | Gen. X+       | 786     |
| Ethnicity | white         | 2168    |
|           | black         | 1091    |
|           | hisp          | 32      |
|           | other         | 5       |
|           | middleEastern | 1       |
| Gender    | man           | 1859    |
|           | woman         | 1412    |
|           | nonBinary     | 26      |

Table 6: Subgroup Counts for Sap et al. 2022. “Support” refers to annotation count.

| Category       | Subgroup             | Support |
|----------------|----------------------|---------|
| Gender         | Female               | 8011    |
|                | Male                 | 6832    |
|                | Unknown              | 124     |
|                | Nonbinary            | 93      |
| Ethnicity      | White                | 10699   |
|                | Black                | 1940    |
|                | Asian                | 862     |
|                | Multiracial          | 664     |
|                | Hispanic             | 418     |
|                | Other                | 247     |
|                | Unknown              | 230     |
| Age            | Under 18             | 3       |
|                | 18 - 24              | 1711    |
|                | 25 - 34              | 5969    |
|                | 35 - 44              | 3785    |
|                | 45 - 54              | 1977    |
|                | 55 - 64              | 1101    |
|                | 65 or older          | 471     |
| Education      | N/A                  | 43      |
|                | No high school       | 71      |
|                | High School graduate | 1353    |
|                | College, no degree   | 3090    |
|                | Associate degree     | 1676    |
|                | Bachelor’s degree    | 6071    |
|                | Master’s degree      | 2261    |
|                | Professional degree  | 236     |
|                | Doctoral degree      | 198     |
|                | N/A                  | 71      |
| Sexual Orient. | Other                | 32      |
|                | Heterosexual         | 12391   |
|                | Bisexual             | 1665    |
|                | Homosexual           | 515     |
|                | N/A                  | 302     |
|                | Other                | 131     |

Table 7: Subgroup Counts for the Kumar et al. 2021 dataset. Part 1 of 2. “Support” refers to annotation count.

(see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only 5 – 6 annotations per comment. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.



| Category        | Subgroup          | Support |
|-----------------|-------------------|---------|
| Is Transgender  | No                | 14549   |
|                 | Yes               | 363     |
|                 | N/A               | 148     |
| Politics        | Liberal           | 5997    |
|                 | Independent       | 4143    |
|                 | Conservative      | 3981    |
|                 | N/A               | 609     |
|                 | Other             | 330     |
| Is Parent       | Yes               | 7889    |
|                 | No                | 7015    |
|                 | N/A               | 156     |
| Tech Impact     | Very negative     | 149     |
|                 | Somewhat negative | 1411    |
|                 | Neutral           | 1905    |
|                 | Somewhat positive | 7690    |
|                 | Very positive     | 3905    |
| Tox Problem     | Never             | 869     |
|                 | Rarely            | 2485    |
|                 | Occasionally      | 5307    |
|                 | Frequently        | 4441    |
|                 | Very Frequently   | 1958    |
| Rel Important   | No                | 4735    |
|                 | Not very          | 1808    |
|                 | Somewhat          | 3548    |
|                 | Very              | 4741    |
|                 | N/A               | 228     |
| Seen Toxicity   | True              | 11505   |
|                 | False             | 3555    |
| Has Been Target | False             | 10741   |
|                 | True              | 4319    |

Table 8: Subgroup Counts for Kumar et al. 2021 dataset. Part 2 of 2. “Support” refers to annotation count.

### C.3 P-VALUE ESTIMATION

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized partitions and compare the observed apunim against this distribution using a one-sample Student- $t$  test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$\begin{aligned}
 H_0 : \text{apunim}_{\text{rand}}(d, g) &= \text{apunim}(d, g) \\
 H_a : \text{apunim}_{\text{rand}}(d, g) &\neq \text{apunim}(d, g)
 \end{aligned} \tag{9}$$

where  $\text{apunim}_{\text{rand}}(d, g)$  denotes the average apunim values obtained from  $t$  randomized annotator partitions.

We account for the presence of multiple subgroups in each PC dimension using a multiple-hypothesis correction. Since our test is parametric (§??), it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per PC group to estimate nDFUs in Equations 5-7, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3). We discuss condition (2) in App. ??. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific dimension. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method Holm (1979). The test is parameterized by the confidence level of our test, which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses Chen et al. (2017)).

## D SOFTWARE

### D.1 INSTALLATION AND USAGE

We release the `apunim` package, which is a Python library that implements both the `nDFU` and `apunim` calculations, as well as its statistical test. The software is available on PyPi. We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (<https://anonymous.4open.science/r/apunim-page>).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project’s repository (<https://anonymous.4open.science/r/aposteriori-unimodality-E8C1>). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy`, `scipy`, `statsmodels`).

### D.2 IMPLEMENTATION DETAILS

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple PC dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions  $\tilde{r}_i$ , which are originally calculated to obtain the apriori polarization values (see Eq. 5), for the p-value calculations (§??). This means that in our current implementation, the number of groups for estimating apriori polarization ( $\tilde{r}_i$  in Eq. 5), and the number of apunim values used in the p-value calculation have to be the same.

## E REPLICATION DETAILS

We use version 1.0.6 of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Eq. 5) and the p-value estimation (§??), and set a confidence level of  $\alpha = 0.05$ .

While the library efficiently handles large numbers of annotators through vectorized operations, performance degrades significantly when processing a high volume of comments, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §3 complete within a minute, the Kumar dataset necessitates approximately one hour of computation using commercial hardware. Furthermore, the annotator variance analysis detailed in App. ?? required over ten hours. While this latter time requirement is anticipated due to the inherently demanding nature and non-standard application of our library, it demonstrates the existence of a potential scaling bottleneck. The same applies to a lesser extent for the subsampling experiments on the DICES datasets, taking collectively three hours of computation.

We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each PC dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each PC subgroup, or running the different experiments in parallel

(indeed, we use the `gnu-parallel` package Tange (2025) for our own experiments). We also note that the Kumar dataset also requires substantial RAM resources, estimated around 42–46GB of dedicated RAM should the whole dataset be considered.

Finally, we note that in Fig. 10, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 73 annotators—see App. B).

## F SYNTHETIC EXPERIMENTS

This appendix details the synthetic experiments used to illustrate the functionality and behavior of nDFU. In all simulations, the bin size is set to  $N_{bins} = 10$ .

**Disagreement vs. polarization** To explore the relationship between annotation polarization and disagreement in Fig. 1, we constructed a  $2 \times 2$  matrix of synthetic datasets, each drawing from a pool of  $N_{annotators} = 100$ . Polarization is simulated by defining a minority group, which constitutes 20% of the total annotators ( $n_{special} = 20$ ). These minority annotators are assigned a targeted mean shift ( $mean\_shift = 1$ ) from the base distribution. The base variance for these distributions is set at  $variance = 0.3$ . Different types of distributions (unimodal, bimodal, multimodal, uniform) are achieved by controlling the `base_means` and applying variance multipliers (e.g.,  $variance \times 3$  for multimodal).

**Single-comment polarization** We simulate the distribution of toxicity annotations in Fig. ?? across two demographic groups (Men and Women). The distribution of Men’s scores is modeled using a truncated normal distribution with  $loc = 2$  and  $scale = 1.3$ , while Women’s scores are modeled with  $loc = 8$  and  $scale = 1.3$ . The combined dataset, representing all annotations, is used to demonstrate the impact of population weighting on the overall nDFU score. The sample size for each group in this visualization is  $N_{annotators} = 50$ .

**Discussion-level polarization** We examine nDFU scores for three specific textual examples presented in Fig. 3. For the individual comments,  $N_{annotators} = 200$  for each group. In the first case (misogynist comment), Men’s scores are modeled with  $loc = 6$ , while Women’s scores are modeled with  $loc = 2$ . In the second case (misandrist comment), the roles are reversed. The third visualization combines distributions for men and women derived from the respective single-comment distributions.

## G HOW DOES INPOL LOOK LIKE?

Table 9 shows a random sample of comments taken from the Sap dataset and their respective InPol values (§4.2). Note that explicitly toxic or racist comments tend to have InPol values of zero, alongside clearly unproblematic comments. Inversely, more ambiguous and less comprehensible comments tend to have larger InPol values. This pattern may suggest that InPol may be tied to the overall ambiguity of the statements, especially in cases where comments may either be interpreted as very toxic, or non-toxic, which agrees with the overall goal of nDFU (§2.2). We include the InPol values for each individual comment for all four datasets in the supplementary material.

## H ABLATION EXPERIMENTS

### H.1 WHY DO STATISTICAL METHODS FAIL?

In §5.2 we mentioned that the underperformance of the  $\chi^2$  and GMM methodologies is likely caused by opposing polarization effects. An ablation experiment with a much simpler setup, where annotators of the same group just shift their opinions by a range in  $[-\frac{\delta}{2}, \frac{\delta}{2}]$  in all comments shows that these methods do indeed show excellent performance. This is expected, since they rely on consistent fixed means effects, which is disrupted in our original formulation of the problem. Thus, while these methods perform admirably in ideal conditions, they may completely fail at realistic annotation datasets.

| Comment                                                                                                                                                                                                                                                              | InPol |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| The minks mentioned in the headline are indeed "rising" from where they were based, but that is due to them being buried in shallow mass graves (Africa) and filling with gas that causes them to rise out. Rest assured, they're still very much dead, thankfully!. | 1     |
| There should be such thing as a bitch resistance for us women                                                                                                                                                                                                        | 1     |
| There's a big difference between being a bitch and being a woman                                                                                                                                                                                                     | 1     |
| i just watched a video of @joisnotonfire being an annoying hoe on skype i'm laughing                                                                                                                                                                                 | 1     |
| niggas love to commit crime                                                                                                                                                                                                                                          | 1     |
| I swear to God above I love ya and when ya be with this brotha, I swear I love you my bro as long as you not a nigga                                                                                                                                                 | 0.5   |
| I will become a cold ass bitch because of my treatment                                                                                                                                                                                                               | 0.5   |
| I would happily lay out a nigga                                                                                                                                                                                                                                      | 0.5   |
| Shoot yourself in the head you hypersensitive fuckin bitch.                                                                                                                                                                                                          | 0.5   |
| not all women are bitches, but all bitches are women                                                                                                                                                                                                                 | 0.5   |
| "you ugly bitch!!!"- that seems to be my real name these days"                                                                                                                                                                                                       | 0     |
| "I am happy to report that drowning does kill this demonic entity. Oh holy are the waters of Chernarus!"" Omg, DED ;)"                                                                                                                                               | 0     |
| "Let them praise the name of the LORD: for his name alone is excellent; his glory is above the earth and heaven."" Psalms 148:13 #Bible"                                                                                                                             | 0     |
| someone plz get this nigger some help. He's playing up like mad                                                                                                                                                                                                      | 0     |
| stop talking you fucking nigger !                                                                                                                                                                                                                                    | 0     |

Table 9: Sample of comments from the Sap dataset and their InPol. Note that unambiguously toxic comments can be assigned with zero InPol, while more nuanced cases tend to have higher values.

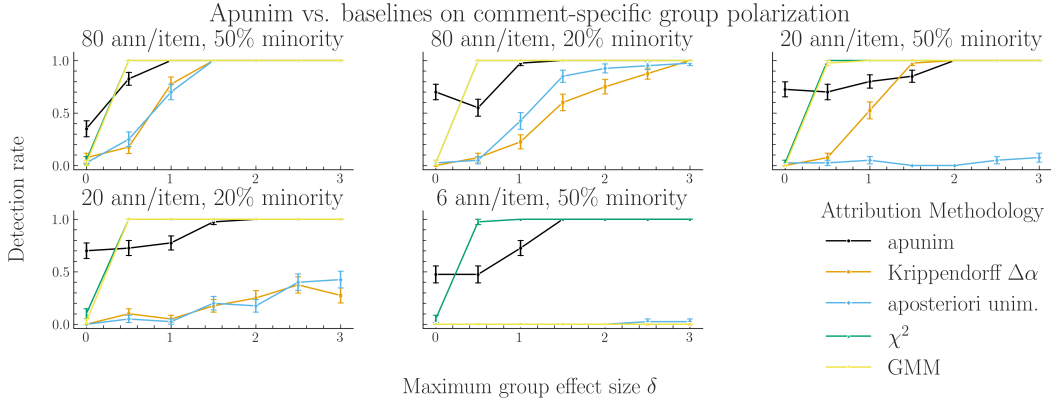


Figure 8: Polarization group attribution on randomly generated, synthetic distributions. Bars indicate standard error. For each experiment we assign a number of annotators per item, and split the annotator groups 50-50 or 80-20. The setup is much more simplified, relying on fixed-mean differences instead of the setup used in §5.2.

## H.2 HOW DOES DATASET SIZE INFLUENCE APUNIM?

Since our approach was designed for the small-scale datasets common in NLP, we were cautious when testing it on the massive Kumar dataset. Because of the data volume, we analyzed various smaller samples (1,000, 10,000, and 30,000 comments), shown in Table 10 (rows with more than one N/A values removed). As sample size increased, the statistical metrics (apunim values) consistently shrank, and we found no statistically significant effects in the larger samples.

The test is overly conservative, meaning it almost always suggests there is no significant effect, even when an effect likely exists in reality. This inability to detect a real signal is a Type II Error (a false negative). This suggests that our test is statistically weak or “underpowered” in massive settings.

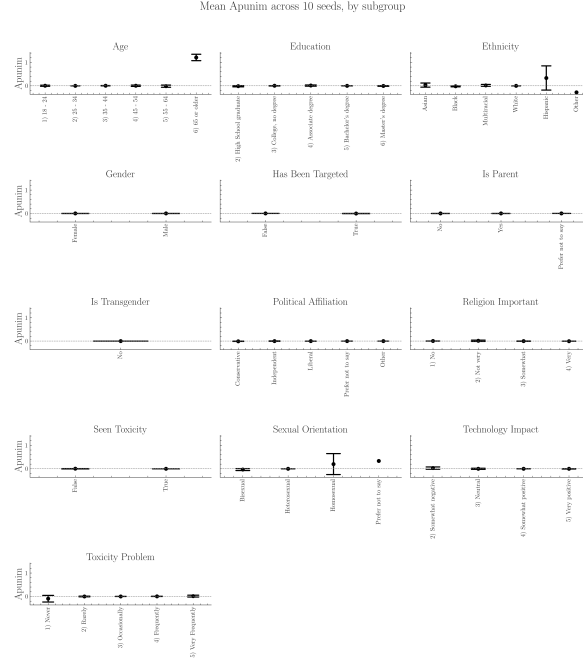


Figure 9: Apunim values on the Kumar dataset ( $n_{samples} = 3000$ ) across ten runs with different subsets. Black bars indicate two SDs. PC dimensions with many groups are more volatile, although all dimensions broadly keep the same apunim sign across subsets.

Since our methodology is built for the small-scale NLP datasets that dominate the research field, we deem this limitation acceptable for the scope of our work.

### H.3 HOW DOES SAMPLE SELECTION INFLUENCE APUNIM?

Astute readers may notice that some variability does exist across differently sized samples for the same examined groups, where for example the sign of the apunim value may switch. We note that the statistically significant values (bold rows) change signs only when the apunim value approaches zero in the largest samples. Thus, the results are not inverted, but rather pushed towards zero.

In order to verify that this variability is caused by the sample size, and not by different samples being drawn, we execute a further ablation study. Specifically, we execute the apunim study on the 3000-sized sample of the dataset, using different subsets of the Kumar dataset. The results can be found in Fig. 9, where we note that variability is not an issue, except for a couple of cases. These cases seem to represent sparse annotator groups, where missing a few comments may genuinely change the value of the apunim metric. These groups however are largely not detected by apunim as significant either way (see §6).

## I THE IDEAL SUBJECTIVE DATASET

We briefly discuss the type of data most suited for our proposed methodology, and connect it with current NLP datasets.

**How many annotators?** Our methodology does not directly require numerous annotators overall, only a sufficient number per item—indeed, statistically significant results can be obtained with as little as four annotations (see Sap-AA; Table 11). A large annotator pool however is required to get sufficient coverage on the examined groups. More annotators allow both higher fidelity of each PC dimension, and more such dimensions.

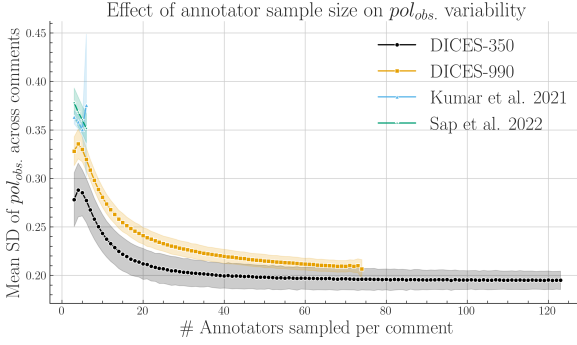


Figure 10: SD of  $pol_{obs}$  across comments, averaged across 1000 runs (Eq. 7). Shaded areas represent two SDs. Pol. estimates benefit from more annotators up to  $n = 20$ , at which point they begin to plateau.

**How many annotators per item?** We investigate observed polarization against varying numbers of annotators. Specifically, for every comment, we repeatedly sampled  $n$  annotators with replacement, calculated  $pol_{obs}$  for each sample and measured SD, averaging it across 1000 same-sized samples. As shown in Fig. 10,  $pol_{obs}$  exhibits a clear, inverse relationship with the number of annotators, attributed to better histogram estimation provided by nDFU (§??), though the rate of gain substantially diminishes after 20 annotators. Interestingly, some variability persists even with a very high annotator count.

**How many items?** Our methodology is designed to work with smaller datasets common in NLP. Three of the four datasets in the study feature low (300–600) item counts, while the Kumar dataset has been subsampled to 3% of its original size. Large datasets should be used only to provide larger coverage on the target domain, since a much better representation of individual groups can be achieved with more annotators per item, rather than more items. Smaller datasets may also be more practical for social science analysis Rossi et al. (2024).

**Filtering datasets with desirable levels of subjectivity** In order to investigate whether the numerous annotators per comments can explain the lack of InPol in the DICES datasets (§4.3) we gradually subsample the annotators for each comment randomly with replacement, starting from five annotators per comment, and scaling up to the full complement of annotators in the dataset with a step of two. For each such step, we choose ten annotator samples and compute the maximum InPol across all runs. Even with annotator counts similar to the Kumar and DICES datasets, InPol remains zero, suggesting that the absence of InPol can not be attributed to annotator number alone—rather, it is plausibly a pattern attributable to the DICES datasets themselves. This kind of analysis may be used in the exploratory phase of dataset selection in order to filter out datasets without enough group-specific subjectivity.

## J FULL RESULTS

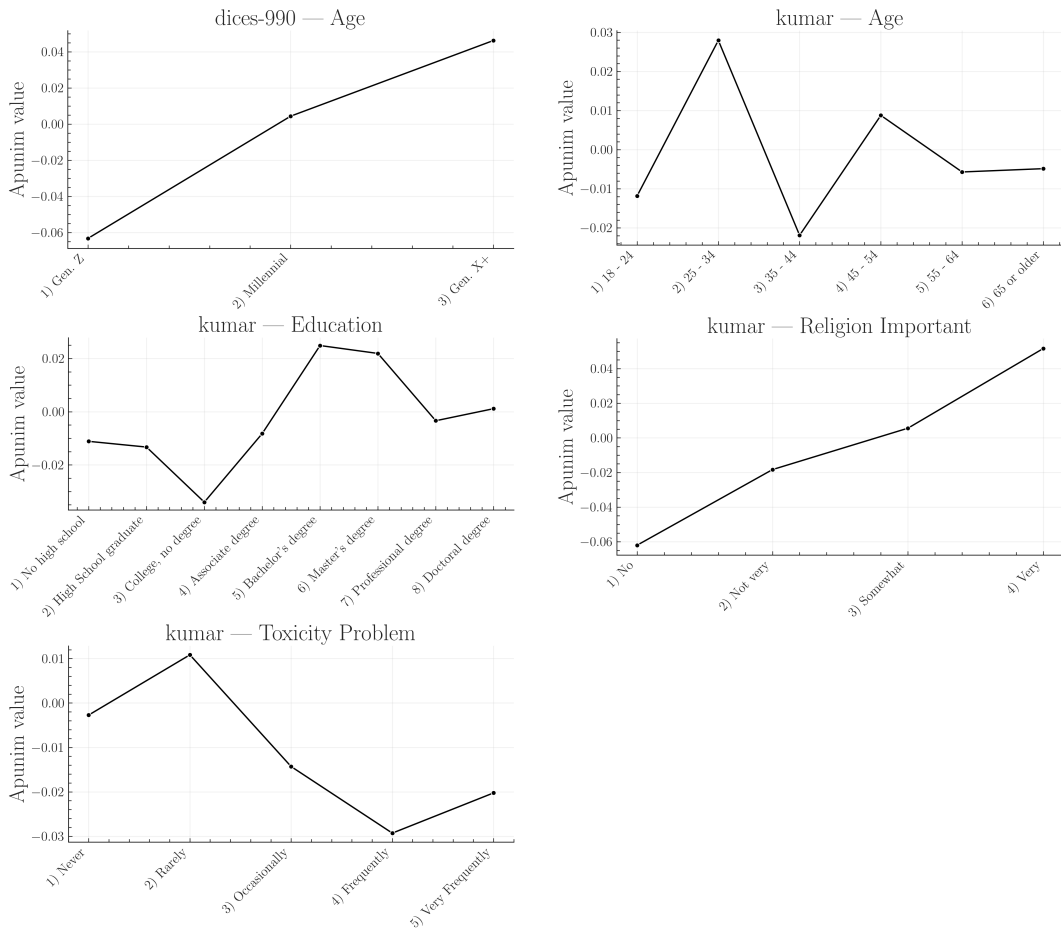


Figure 11: Apunim values expanded for each ordinal PC dimension. This figure is complementary to Fig. 5. The values in the graphs were extracted from the Tables of App. J.

| PC                           | 1000  | 10000 | 30000 |
|------------------------------|-------|-------|-------|
| <b>Age</b>                   |       |       |       |
| 8 - 24                       | -.002 | -.037 | -.016 |
| 25 - 34                      | .045  | 0     | 0     |
| 35 - 44                      | -.043 | -.001 | -.001 |
| 45 - 54                      | .001  | .013  | .003  |
| 55 - 64                      | .002  | -.057 | .001  |
| 65 or older                  | -.007 | N/A   | .948  |
| <b>Education</b>             |       |       |       |
| High School graduate         | -.023 | -.005 | .014  |
| <b>College, no degree</b>    | -.045 | .010  | .006  |
| Associate degree             | .001  | .005  | .009  |
| Bachelor's degree            | .029  | -.001 | -.000 |
| Master's degree              | .024  | .004  | -.000 |
| <b>Ethnicity</b>             |       |       |       |
| Asian                        | -.012 | .011  | .007  |
| Black                        | .036  | -.004 | -.012 |
| Hispanic                     | .004  | 0     | -.099 |
| Multiracial                  | .008  | 0     | 0     |
| Other                        | -.008 | N/A   | -.306 |
| <b>White</b>                 | -.023 | 0     | 0     |
| <b>Gender</b>                |       |       |       |
| Female                       | -.006 | .002  | .001  |
| Male                         | .040  | -.002 | -.001 |
| <b>Has Been Targeted</b>     |       |       |       |
| <b>False</b>                 | -.059 | 0     | 0     |
| True                         | .017  | -.003 | -.002 |
| <b>Is Parent</b>             |       |       |       |
| <b>No</b>                    | -.084 | .001  | .001  |
| N/A                          | .004  | 1.000 | 0     |
| <b>Yes</b>                   | .071  | 0     | 0     |
| <b>Is Transgender</b>        |       |       |       |
| <b>No</b>                    | -.102 | 0     | 0     |
| <b>Political Affiliation</b> |       |       |       |
| Conservative                 | .056  | -.003 | -.002 |
| Independent                  | -.003 | .001  | .002  |
| Liberal                      | -.026 | .001  | .001  |
| Other                        | -.003 | 0     | 0     |
| <b>Religion Important</b>    |       |       |       |
| <b>No</b>                    | -.071 | .002  | .001  |
| Not very                     | -.013 | .010  | .006  |
| Somewhat                     | .022  | .003  | -.002 |
| <b>Very</b>                  | .043  | -.003 | -.002 |
| <b>Seen Toxicity</b>         |       |       |       |
| <b>False</b>                 | .025  | -.001 | -.001 |
| <b>True</b>                  | -.082 | .001  | 0     |
| <b>Sexual Orientation</b>    |       |       |       |
| Bisexual                     | -.003 | -.040 | -.013 |
| <b>Heterosexual</b>          | -.067 | 0     | 0     |
| Homosexual                   | -.012 | 0     | .053  |
| <b>Technology Impact</b>     |       |       |       |
| Somewhat negative            | -.008 | 0     | .003  |
| Neutral                      | .017  | -.014 | 0     |
| Somewhat positive            | -.053 | 0     | .001  |
| Very positive                | .027  | -.004 | -.003 |
| <b>Toxicity Problem</b>      |       |       |       |
| Never                        | -.008 | -.037 | -.081 |
| Rarely                       | .004  | .003  | -.001 |
| Occasionally                 | -.012 | -.003 | -.001 |
| Frequently                   | -.032 | -.001 | 0     |
| Very Frequently              | -.017 | .030  | .015  |

Table 10: Comparison of apunim values across different sample sizes (1k, 10k, 30k) for the Kumar dataset. Bold rows show statistically significant results in the 3k sample used in the main text.



| Group                  | apunim | pvalue   | support |
|------------------------|--------|----------|---------|
| <b>Kumar</b>           |        |          |         |
| Age=18-24              | -.0118 | .4822    | 1431    |
| Age=25-34              | .0279  | .1185    | 4888    |
| Age=35-44              | -.0219 | .1733    | 3102    |
| Age=45-54              | .0088  | .5394    | 1668    |
| Age=55-64              | -.0057 | .6592    | 938     |
| Age=65 or older        | -.0049 | .6747    | 409     |
| Age=N/A                | .0000  | N/A      | 37      |
| Age=Under 18           | .0000  | N/A      | 2       |
| Ed=No HS               | -.0111 | .5278    | 60      |
| Ed=High School         | -.0133 | .3461    | 1123    |
| Ed=College             | -.0340 | .0189    | 2508    |
| Ed=Associate           | -.0082 | .5963    | 1362    |
| Ed=Bachelor's          | .0249  | .1091    | 5007    |
| Ed=Master's            | .0219  | .1091    | 1982    |
| Ed=Professional        | -.0034 | .7459    | 192     |
| Ed=Doctoral            | .0011  | .8651    | 164     |
| Ed=Other               | .0000  | N/A      | 28      |
| Ed=N/A                 | .0000  | N/A      | 63      |
| Eth=Asian              | -.0019 | .9332    | 720     |
| Eth=Black              | .0143  | .7070    | 1671    |
| Eth=Hispanic           | -.0075 | .9092    | 352     |
| Eth=Multiracial        | -.0026 | .9332    | 540     |
| Eth=Other              | -.0007 | .9332    | 204     |
| Eth=Unknown            | -.0063 | .9092    | 189     |
| Eth=White              | -.0291 | .0416    | 6619    |
| Gen=Female             | -.0245 | .0720    | 6141    |
| Gen=Male               | .0261  | .0720    | 5479    |
| Gen=Nonbinary          | -.0014 | .9436    | 81      |
| Gen=Unknown            | -.0008 | .9436    | 104     |
| Targeted=False         | -.0482 | 1.04e-05 | 6632    |
| Targeted=True          | .0146  | .2310    | 3643    |
| IsParent=No            | -.0686 | 2.39e-08 | 5514    |
| IsParent=N/A           | .0017  | .8913    | 130     |
| IsParent=Yes           | .0490  | 2.85e-05 | 6216    |
| IsTrans=No             | -.0788 | 7.47e-05 | 1697    |
| IsTrans=N/A            | -.0007 | .9483    | 128     |
| IsTrans=Yes            | -.0073 | .7711    | 340     |
| Pol=Conservative       | .0269  | .0534    | 3386    |
| Pol=Independent        | -.0201 | .1371    | 3432    |
| Pol=Liberal            | -.0278 | .0534    | 4945    |
| Pol=Other              | -.0132 | .2947    | 271     |
| Pol=N/A                | .0034  | .6523    | 511     |
| RelImp=No              | -.0620 | 2.96e-07 | 3864    |
| RelImp=Not very        | -.0183 | .1039    | 1493    |
| Ed=Somewhat            | .0056  | .6169    | 3012    |
| RelImp=Very            | .0516  | 4.58e-05 | 4062    |
| RelImp=N/A             | .0066  | .2968    | 194     |
| SeenTox=False          | .0350  | .0029    | 3050    |
| SeenTox=True           | -.0766 | 4.82e-13 | 6360    |
| SexOrient=Bisexual     | .0087  | .7884    | 1481    |
| SexOrient=Heterosexual | -.0500 | 3.22e-05 | 5721    |
| SexOrient=Homosexual   | -.0048 | .7884    | 430     |
| SexOrient=Other        | .0009  | .9055    | 115     |
| SexOrient=N/A          | -.0061 | .7884    | 265     |
| TechImp=Very neg       | .0000  | N/A      | 118     |
| TechImp=Somewhat neg   | -.0085 | .4097    | 1179    |
| TechImp=Neutral        | .0085  | .4097    | 1603    |
| TechImp=Somewhat       | -.0285 | .0691    | 6040    |
| TechImp=Very pos       | .0105  | .4097    | 3315    |
| ToxProblem=Never       | -.0027 | .8071    | 741     |
| ToxProblem=Rarely      | .0108  | .3965    | 2087    |
| ToxProblem=Occ         | -.0143 | .3809    | 4347    |
| ToxProblem=Freq        | -.0293 | .0507    | 3740    |
| ToxProblem=Very        | -.0202 | .1404    | 1625    |

Table 11: Apunim results for the Kumar dataset. \* denotes  $p < 0.10$ , \*\* denotes  $p < 0.05$ , and \*\*\* denotes  $p < 0.01$ . Rows shaded in gray indicate a negative apunim value.

| Group           | apunim | pvalue | support |
|-----------------|--------|--------|---------|
| Sap             |        |        |         |
| Age=GenX+       | -.066  | .708   | 271     |
| Age=Millennial  | .007   | .744   | 1886    |
| Ethnicity=black | .042   | 4.000  | 4       |
| Ethnicity=white | -.000  | .998   | 1896    |
| Gender=Man      | -.091  | .000   | 1439    |
| Gender=Woman    | .244   | .000   | 877     |

Table 12: Apunim results for the Sap Dataset. \* denotes  $p < 0.10$ , \*\* denotes  $p < 0.05$ , and \*\*\* denotes  $p < 0.01$ . Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

| Group            | apunim | pvalue   | support |
|------------------|--------|----------|---------|
| DICES-350        |        |          |         |
| Age Gen. Z       | .007   | .539     | 17528   |
| Age Millennial   | .006   | .539     | 11268   |
| Age Gen. X+      | -.019  | .166     | 9703    |
| Ed=College +     | -.012  | .234     | 23475   |
| Ed=High school   | -.015  | .186     | 12833   |
| Ed=Other         | .021   | .025     | 2191    |
| Gender Man       | .019   | .058     | 19093   |
| Gender Woman     | -.021  | .058     | 19406   |
| Race African Am. | .043   | 2.41e-05 | 9077    |
| Race Asian       | -.066  | 1.16e-10 | 8138    |
| Race Latino      | .003   | .920     | 6886    |
| Race Multiracial | .000   | .980     | 5008    |
| Race White       | .020   | .067     | 9390    |

Table 13: Apunim results for the DICES-350 Dataset. \* denotes  $p < 0.10$ , \*\* denotes  $p < 0.05$ , and \*\*\* denotes  $p < 0.01$ . Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

| Group            | apunim | pvalue    | support |
|------------------|--------|-----------|---------|
| DICES-990        |        |           |         |
| Age Gen. Z       | -.063  | 3.05e-25  | 13741   |
| Age Millennial   | .004   | .495      | 38436   |
| Age Gen. X+      | .046   | 7.17e-15  | 14384   |
| Ed=College +     | -.009  | .163      | 59142   |
| Ed=High school   | -.075  | 5.80e-42  | 7419    |
| Gender Man       | -.027  | 3.81e-05  | 32494   |
| Gender Woman     | .025   | .00015    | 33471   |
| Race African Am. | -.091  | 1.61e-64  | 5810    |
| Race Asian       | -.011  | .080      | 18428   |
| Race Latino      | .133   | 7.57e-112 | 6610    |
| Race Other       | -.075  | 1.51e-30  | 24321   |
| Race White       | .120   | 6.03e-81  | 11392   |

Table 14: Apunim results for the DICES-990 Dataset. \* denotes  $p < 0.10$ , \*\* denotes  $p < 0.05$ , and \*\*\* denotes  $p < 0.01$ . Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.